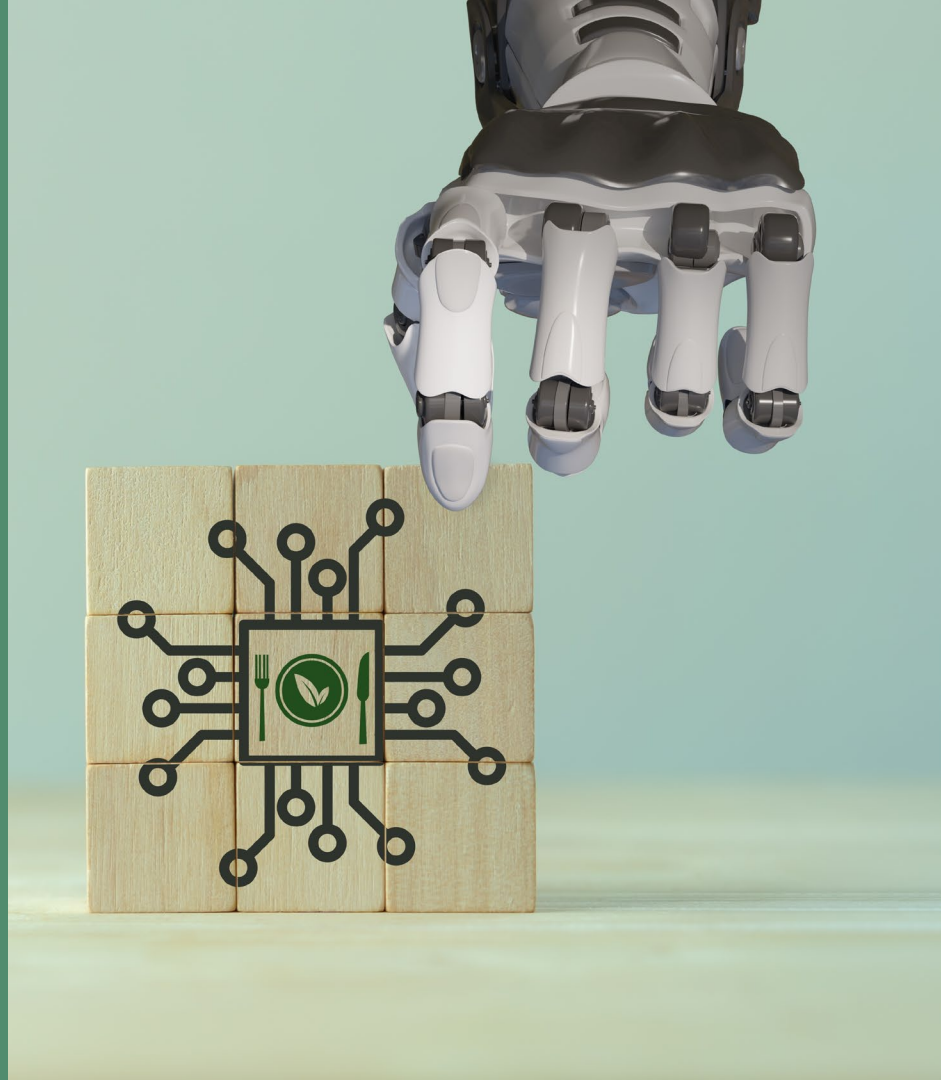
AI in Food Safety

2026 FOOD LABEL CONFERENCE

Claire Zoellner, PhD

June 1, 2026

FoodDS®





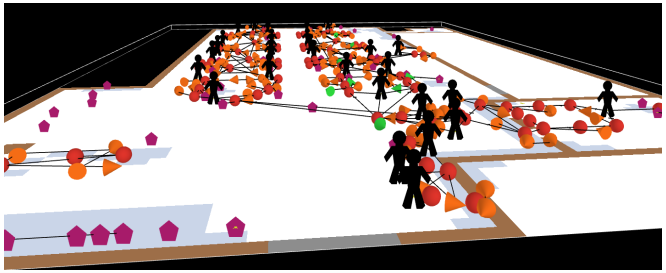
...to harness AI's capacity to ***improve quality control processes*** *and* ***drive product innovation****, and ...* ***enhance employee productivity and digital competencies*** *without reducing employment.*

Goldman Sachs, 2024

LISTERIA CONTROL: AGENT-BASED MODELING & DIGITAL TWIN

- Recreate a food processing facility *in silico*
- Simulate production over time to observe *Listeria* introduction, transmission, removal and persistence
- Supplement historical sampling data with model predictions to
  - Optimize EMP resources
  - Visualize contamination pathways
  - Test different strategies





› Sci Rep. 2019 Jan 24;9(1):495. doi: 10.1038/s41598-018-36654-z.

EnABLe: An agent-based model to understand Listeria dynamics in food processing facilities

Claire Zoellner [1], Rachel Jennings [2], Martin Wiedmann [3], Renata Ivanek [2]

› Appl Environ Microbiol. 2021 Aug 18;AEM0079921. doi: 10.1128/AEM.00799-21.
Online ahead of print.

Using *in silico* models for design and optimization of science-based *Listeria* environmental monitoring programs in fresh-cut produce facilities

Genevieve Sullivan [1 2], Claire Zoellner [3], Martin Wiedmann [1], Renata Ivanek [2]

FOOD SAFETY TRANSFORMATION BRINGS AI-READINESS



2004

iDecisionSciences (iDS)

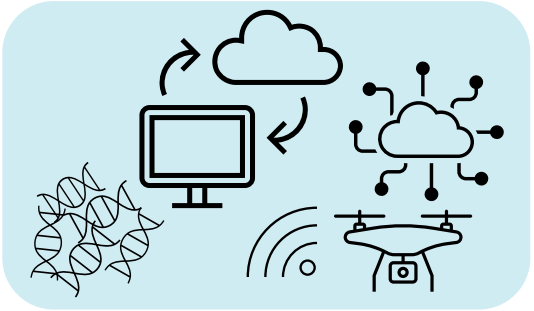
a Seattle-based consulting agency serving the fresh produce industry



2013

iFoodDecisionSciences, Inc.

software provider managing data requirements of food safety programs



2020

iFoodDS

supply chain platform for **safety**, **traceability**, and **quality management**

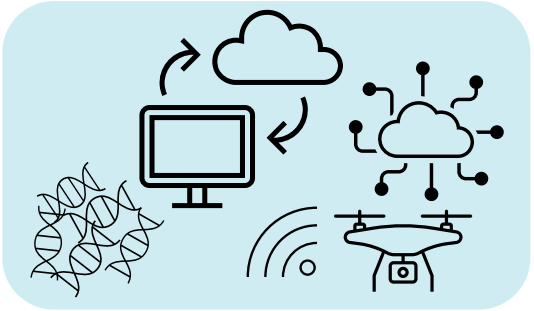
FOOD SAFETY TRANSFORMATION BRINGS AI-READINESS



2004



2013



2020

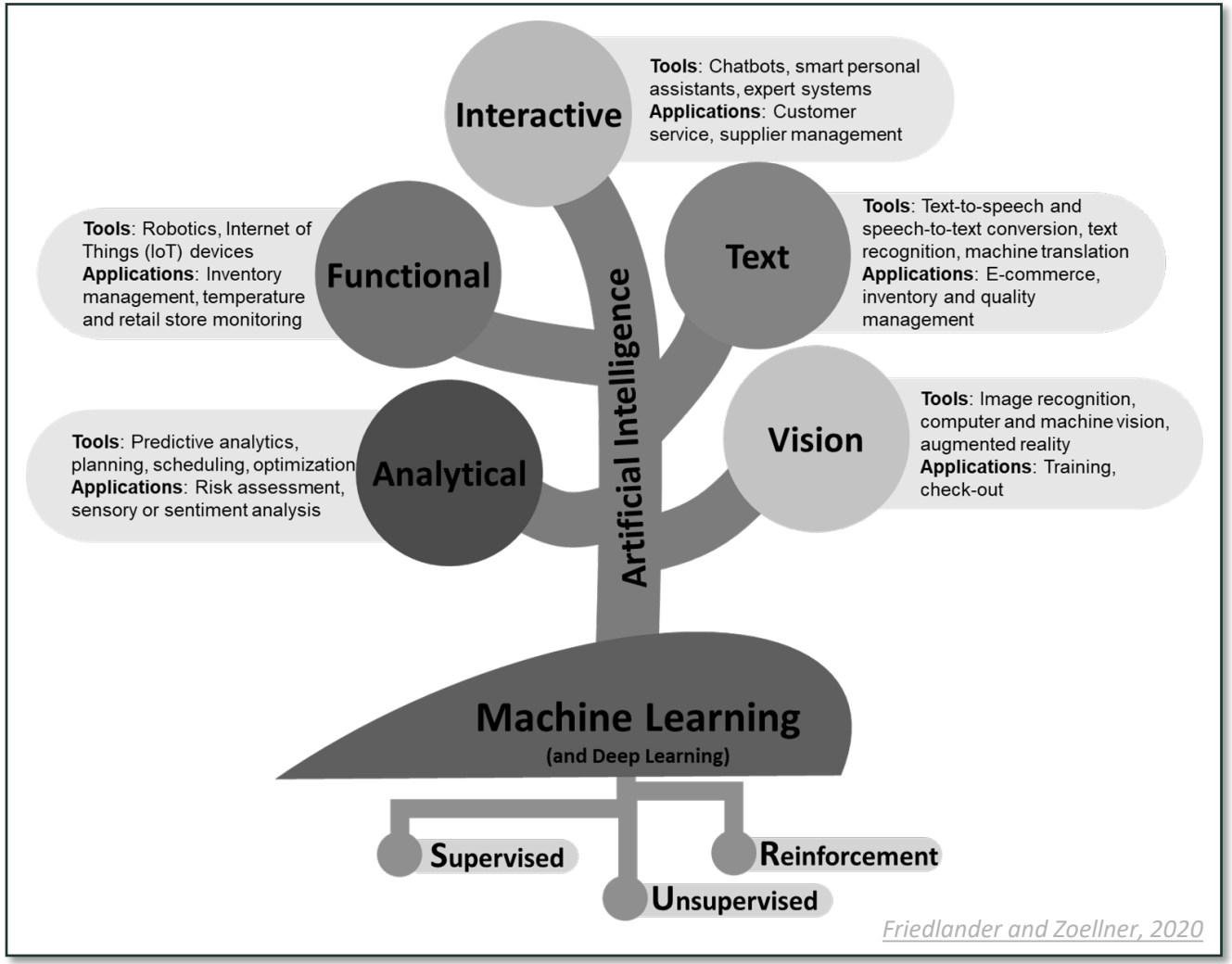
Healthy People 2020

FDA FOOD SAFETY MODERNIZATION ACT

NEW ERA OF SMARTER FOOD SAFETY
FDA's Blueprint for the Future

Healthy People 2030

2020



Friedlander and Zoellner, 2020

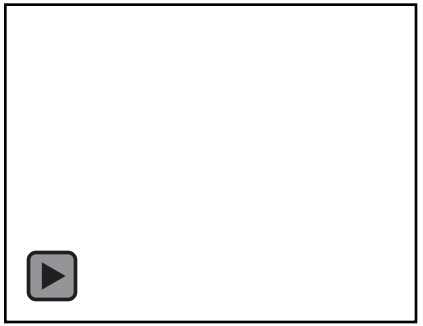
AI IN FOOD SAFETY: REFINE RISK COMMUNICATION

Using chatbots and Augmented Reality for upskilling and training on continuous improvement

KEY APPLICATIONS

- Translate or simplify complex topics and large amounts of information
  - Practice messaging for difficult conversations
  - Write food safety plans
  - Identify relevant regulatory requirements
- Visualize microbial and chemical contamination sources and transmission routes – employee training







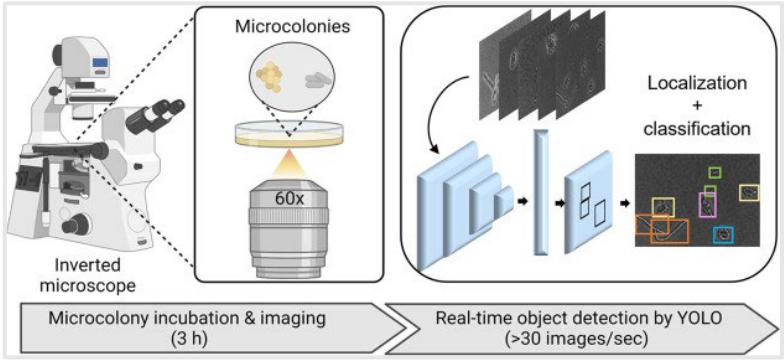


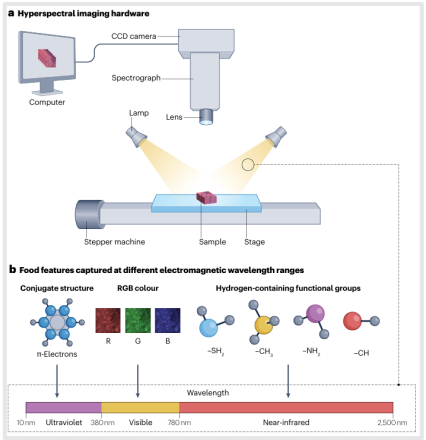
AI IN FOOD SAFETY: FACILITATE RISK ASSESSMENT

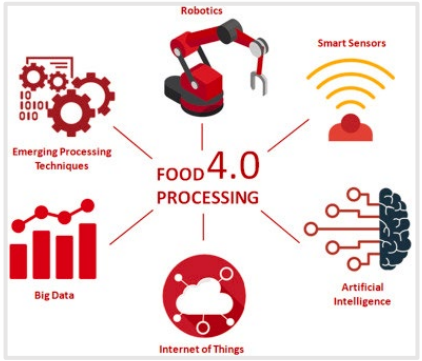
Combining microbial data with IoT, computer vision, and/or AI/ML to detect and interpret new risks

KEY APPLICATIONS

- Shift from batch testing to continuous and automated sampling
  - Data-driven, non-destructive methods
- Add environmental context of microbial data; forecast contamination events
  - Real-time monitoring of environment and SOP compliance in facilities
- Ensure food integrity and traceability
  - Automate data capture and sharing







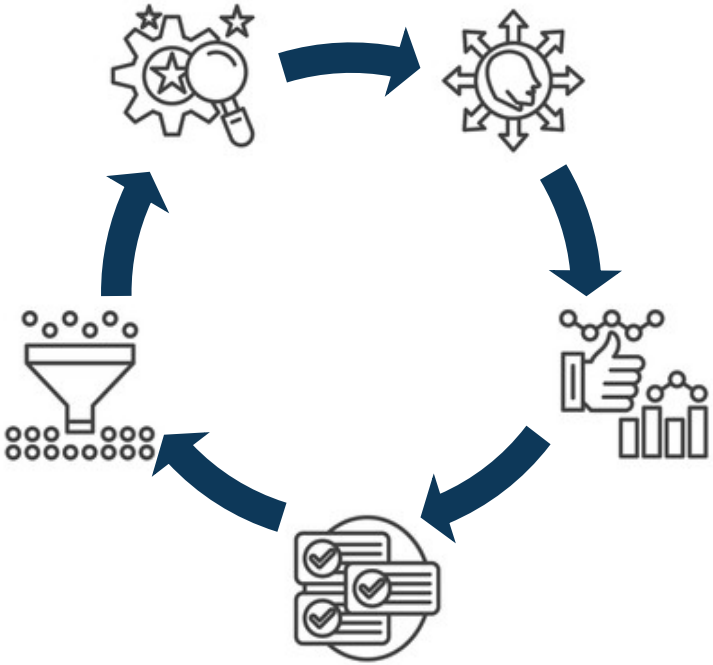
Sun et al., 2024; Qian et al., 2023; Hassoun et al., 2023; Ma et al., 2022

AI IN FOOD SAFETY: ENABLE RISK MANAGEMENT

Supporting and automating decision-making with predictive analytics and/or simulation

KEY APPLICATIONS

- Automate information analysis and recommend actions
  - FDA PREDICT (Predictive Risk-based Evaluation for Dynamic Import Compliance Targeting)
  - FDA Product Tracing System
  - Enterprise Horizon Scanning Platforms



FDA PREDICT; FDA Product Tracing

FOOD SAFETY STILL WORKING TOWARDS AI ADOPTION

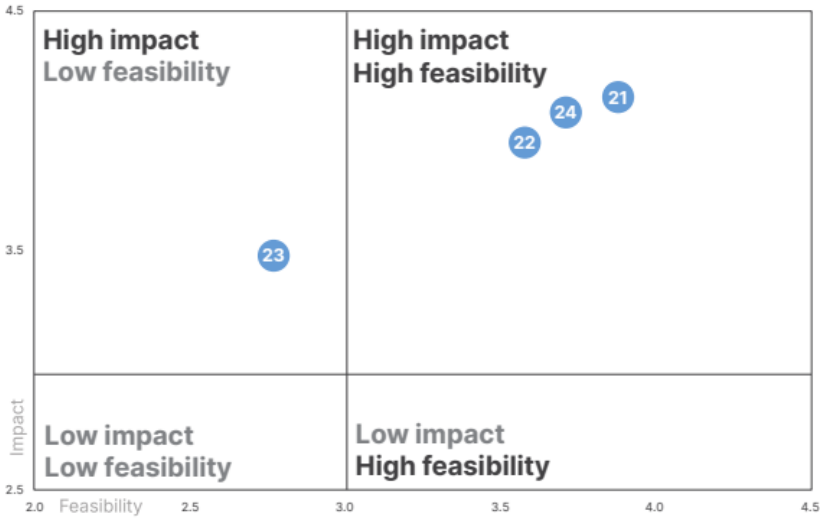
5+ years

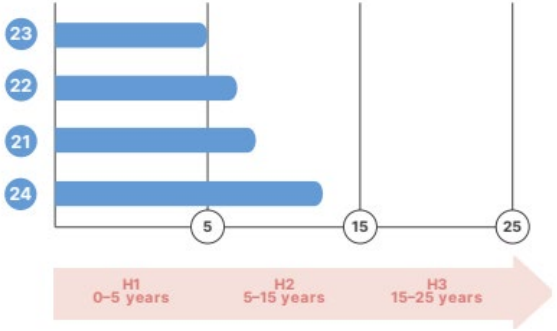
until widespread AI adoption and meaningful impact realized in food safety, according to FAO experts

KEY FINDINGS

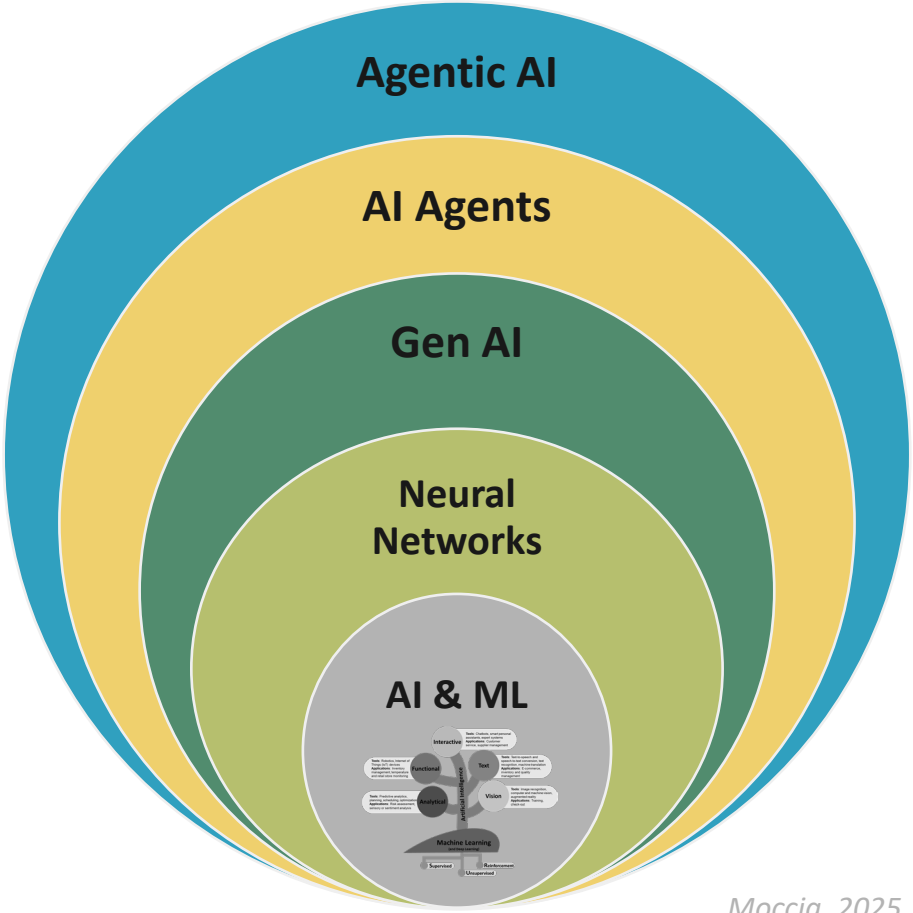
- Digital food twins (23) deemed not very feasible despite high potential impact
- AI (21) and big data + IoT (22) expected to gain momentum in the next 5-year horizon
- Blockchain (24) adoption expected in 15-year timeframe

Mukherjee et al., 2025





2026



Moccia, 2025

Agentic AI
Automate entire processes with AI
Self-improving agents · Agent protocols & marketplaces · Dynamic tooling · Observability

AI Agents
Execute complex tasks autonomously
Memory systems · Goal decomposition · Feedback loops · Human-in-the-loop · Failure recovery · Cost management

Gen AI
Generate content and code at scale
LLMs · RAG · Prompt engineering · Video/image generation · Code generation · Speech interfaces

Neural Networks
Complex pattern detection
CNNs · RNNs · LSTMs · Multimodal generation · Fine-tuning

AI & ML
Turn data into decisions
Supervised learning · Unsupervised learning · Reinforcement learning · NLP · Predictive models

THE OPPORTUNITY

9.9M

Annual cases of foodborne illness in US – top 6 pathogens

CDC, 2025

$36B

Annual economic burden of foodborne illness in US – top 6 pathogens

Hoffman et al., 2025

20.7M

US employment in farms, manufacturing, retail and foodservice sectors

USDA ERS, 2023

58%

Percent of AI users that say they're doing work they couldn't have a year ago

Microsoft 2026 Work Trend Index Annual Report

AI FOR *ENTERPRISE* FOOD SAFETY: PREVENTING INCIDENTS

A new operating model - assisted actions (or inaction) at different levels based on new/early indicators or signals

KEY APPLICATIONS

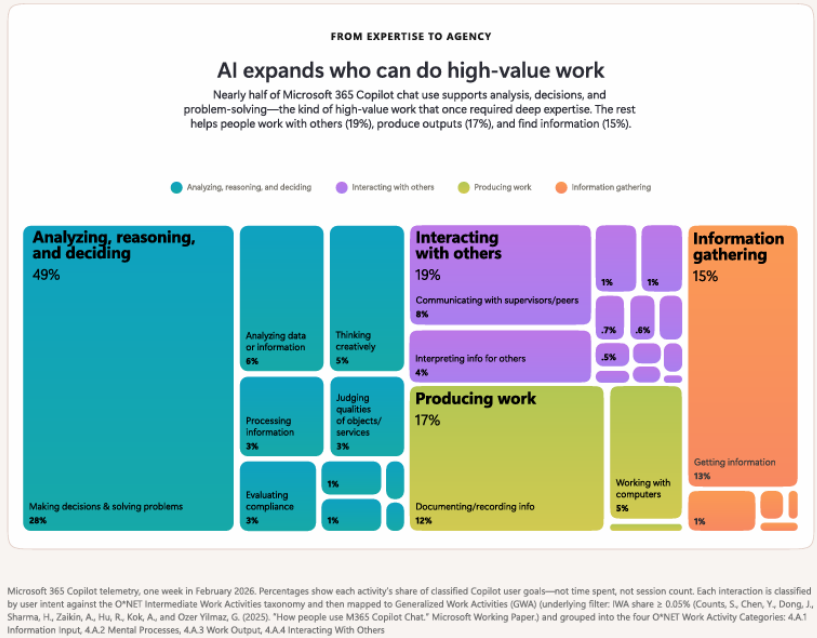
- Scoped AI platforms for your enterprise data
- Skilled AI agents for your workflows, standards, and systems
- Governed and auditable access for your employees

Microsoft Copilot

Claude

Gemini

ChatGPT



Microsoft 2026 Work Trend Index Annual Report

POTENTIAL BARRIERS HAVEN'T CHANGED

Key considerations when adopting AI/ML, including in food safety programs

DATA ACCURACY
Reliable, validated inputs underpin every downstream insight; outputs reviewed for accuracy

DATA PRIVACY & SECURITY
Proprietary information must stay protected; business systems must remain secured against vulnerabilities

REGULATORY RISK
Liability and compliance concerns when incidents occur

ETHICAL CONSIDERATIONS
Transparency, and accountability in model development and outputs; workforce development

ORGANIZATIONAL POLICY
Established internal governance and acceptable-use guardrails

COST
Accounting for infrastructure upgrades, talent acquisition, data fees, and ongoing maintenance investment

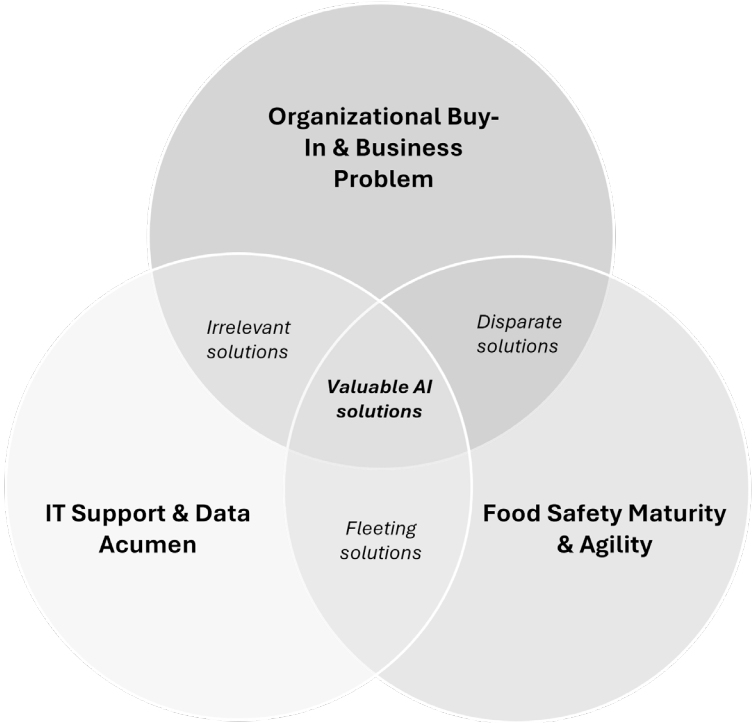
Alexander, et al. 2023

A FRAMEWORK FOR AI READINESS FOR FOOD SAFETY

A survey tool to evaluate technical and cultural readiness for using AI to solve food safety challenges

KEY COMPONENTS

- Align AI + food safety initiatives with strategic company goals
- Building data infrastructure and literacy across teams
- Established food safety processes that can adapt to AI-driven insights



Zoellner & Miller, 2026

HOW TO APPROACH “HUMAN-CENTERED AUTOMATION”

What system tasks should be automated and to what extent?

- Information Acquisition
- Information Analysis
- Decision Selection
- Action Implementation

Parasuraman, et al. 2000

Levels of Automation

High

10. The computer decides everything, acts autonomously, ignoring the human
9. Informs the human only if the computer decides to
8. Informs the human only if asked
7. Executes automatically, then necessarily informs the human
6. Allows the human a restricted time to veto before automatic execution
5. Executes the suggestion if the human approves
4. Suggests one alternative
3. Narrows the selection down to a few alternatives
2. The computer offers a complete set of decision/action alternatives
1. The computer offers no help; the human must take all actions and make decisions

Low

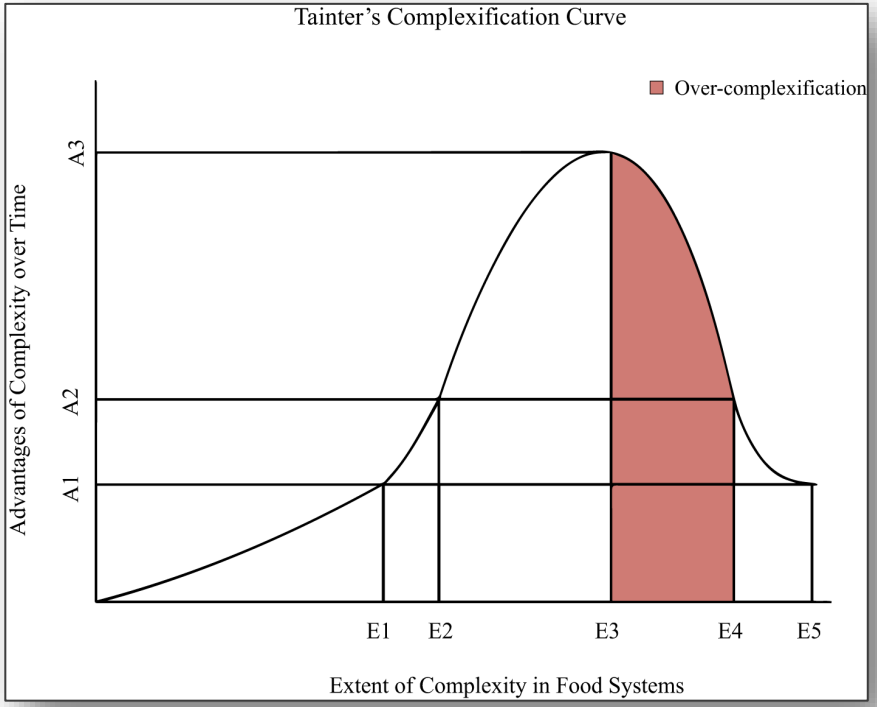
THRESHOLD OF OVER-COMPLEXIFICATION

What value are we bringing to solving the problem and can it be sustained?

While complexification can be an effective problem-solving strategy, over-complexification can be counterproductive

KEY POINTS

- A central indicator of over-complexification is declining return on investment
- Focus on "innovation by subtraction" or technologies that replace complexity



Loring and Sanyal, 2021; Tainter, 1995

OPPORTUNITIES FOR COORDINATION

How to ensure that food safety remains a coordinated effort across the supply chain?

1 ADOPT DATA STANDARDS

- Define standards where needed or adopt existing ones
- Monitor data quality

2 THE RIGHT USE CASES

- Relevant use cases that scale
- Include regulatory and legal
- Share examples of success and short-comings

3 DATA AVAILABILITY

- Use inclusion/ exclusion criteria for data sharing
- Explore use of synthetic data to augment real data

4 VALIDATION STUDIES

- Capture data that can be used prospectively to assess model performance
- Identify metrics
- Iterate using “sandbox” or phased approach

HOW WILL WE MEASURE PROGRESS?

A Vision for Food Safety in 2030

1 Product Traceability

Scan a bag of lettuce and immediately know where it came from and if it's tied to an outbreak

2 Recall Alerts

Receive a text message that says you've purchased something that's been recalled

3 Predictive Detection

AI enables a significant increase in predictive capability of finding contamination sources and contaminated food

4 Global Awareness

Know the potential impact of weather events on the safety of foods growing hundreds of miles away

5 Smart Monitoring

Production environments and inputs are monitored in real-time using sensors to improve root cause analysis and preventive controls

6 Public Health

Reduced rates of foodborne illness (e.g., **11.5** *Salmonella* infections per 100,000 ppl)

FDA New Era of Smarter Food Safety, 2020

TOP TAKEAWAYS

- Food safety transformation supports AI-readiness
- AI is shifting from narrow statistical models and individual tasks to end-to-end decision execution
- The highest-value applications combine AI with food safety context and expertise
- Organizational change and investment matter as much as or more than the technology
- Leadership and collaboration will continue to drive adoption across the food safety community

THANK YOU

Claire Zoellner, PhD

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www.iFoodDS.com

iFoodDS



ABOUT THE SPEAKER

Claire Zoellner, Ph.D., is the Director of Data Science at iFoodDS, a provider of food safety, quality, and traceability software. In her role, she provides leadership for the analytics and data science team as well as guidance on data strategy, exploration of new technologies and methods, and implementation of data science projects. Prior to iFoodDS, Claire worked as a postdoctoral researcher on estimating and mitigating the risk of Listeria in frozen foods, for which she received the IAFP Freezing Research Award. Claire earned her B.S. in Food Science from the University of Illinois and her Ph.D. in Food Science from Cornell University. She was recognized as an emerging leader in food science by IFT in 2019 and remains active in food safety research through publications, service, and collaborations.